

PHYS 201L

University Physics Laboratory I | Fall 2026

Instructor Dr. Edward J. Brash	Meeting Mondays, 6:00–8:45 p.m.
Email edward.brash@cnu.edu	Office Luter 304, or online
Phone 757-594-7451	Course repository github.com/brash99/Phys201L
Office hours MWF 1:00–2:00 p.m., or by appointment	Format In-person laboratory; team notebook submission

Course purpose

This laboratory accompanies PHYS 201, the first semester of calculus-based university physics. The course develops experimental judgment through measurements in mechanics, uncertainty analysis, computational data analysis, graphical communication, and collaborative scientific writing. Computer use is integrated throughout the course through Jupyter notebooks executed in Google Colab and stored in team GitHub repositories.

PHYS 201 lecture is a co-requisite for this laboratory. If you drop or withdraw from the lecture, the Registrar may remove you from the laboratory course.

Learning outcomes

By the end of the course, a successful student should be able to:

- design and carry out reproducible measurements using common mechanics laboratory instruments;
- estimate instrumental and statistical uncertainties and propagate them through calculated quantities;
- compare experimental results with models, accepted values, and conservation laws;
- analyze and visualize data using Python, NumPy, pandas, and Matplotlib in a Jupyter notebook;
- interpret slopes, intercepts, trends, error bars, and fit parameters physically;
- write a clear technical account of experimental methods, evidence, limitations, and conclusions; and
- work responsibly as a laboratory team while documenting individual contributions.

Materials and technology

No separate laboratory textbook is required. OpenStax University Physics, Volume I may be useful for background. Course notebooks, handouts, and resources are published at <https://github.com/brash99/Phys201L>.

Each team needs access to GitHub and Google Colab during laboratory meetings. No local software installation or local repository clone is required.

How the laboratory works

Seven experiments are completed over fourteen laboratory meetings. Each experiment occupies two Monday meetings:

1. During the first meeting, teams review the model and apparatus, configure and calibrate equipment, begin measurements, and check data quality.
2. During the second meeting, teams complete measurements, perform the analysis, create tables and figures, and finish the written notebook.
3. The completed team notebook is due by 8:45 p.m. at the end of the second meeting. Work submitted after the laboratory period requires prior approval or a documented excused absence.

Teams and repository workflow

Each laboratory team maintains one public fork of the course GitHub repository. One student serves as repository manager, adds teammates as collaborators, verifies the final Colab save, and submits the notebook URL through the LMS. All teammates are responsible for the scientific and written content of the submitted notebook.

Do not place student ID numbers, grades, private feedback, or other sensitive information in the public repository. Released laboratory notebooks should not be renamed or moved. New laboratories are obtained with GitHub's Sync fork function.

Reports and effort reports

Team notebook (12% each).

One notebook is submitted per team for each experiment. It must identify all team members and include the team's measurements, calculations, tables, figures, uncertainty analysis, discussion, conclusions, and requested appendices. The saved GitHub version must contain the final executed outputs.

Individual effort report (16% total).

Each student submits an individual effort report for every experiment. The seven reports are equally weighted within this category and are due with the corresponding team notebook. Reports should accurately describe each member's contribution. Effort reports are private and must not be committed to the public team repository.

Assessment

Final grades are based on the seven team notebooks and consistent, candid completion of individual effort reports. A notebook grade may be adjusted for an individual student when instructor observations and effort reports provide clear evidence that the student's contribution differed materially from the team's work.

Category	Calculation	Course weight
Team laboratory notebooks	$7 \times 12\%$	84%
Individual effort reports	7 equally weighted reports	16%

Letter grades

A = 87–100%; A– = 80–86%; B+ = 77–79%; B = 73–76%; B– = 70–72%; C+ = 67–69%; C = 63–66%; C– = 60–62%; D+ = 57–59%; D = 53–56%; D– = 50–52%; F < 50%.

Calendar of experiments

Labor Day classes meet. Fall recess begins after classes on October 9, so there is no laboratory meeting on October 12. Thanksgiving recess begins after classes on November 24.

Lab	Experiment	Meeting dates	Notebook due
1	Density measurements and uncertainties	Aug 24, Aug 31	Aug 31, 8:45 p.m.
2	Air track I: Kinematics	Sep 7, Sep 14	Sep 14, 8:45 p.m.
3	Projectile motion	Sep 21, Sep 28	Sep 28, 8:45 p.m.
4	Air track II: Newton's second law	Oct 5, Oct 19	Oct 19, 8:45 p.m.
5	Air track III: Conservation of energy	Oct 26, Nov 2	Nov 2, 8:45 p.m.
6	Collisions in two dimensions	Nov 9, Nov 16	Nov 16, 8:45 p.m.
7	Moment of inertia	Nov 23, Nov 30	Nov 30, 8:45 p.m.

Attendance, participation, and safety

Attendance is expected at every laboratory meeting because apparatus access, team measurements, and in-person collaboration cannot be reconstructed fully outside the scheduled period. Arrive on time, follow apparatus-specific safety instructions, protect laboratory equipment, and contribute actively to setup, measurement, analysis, and cleanup.

If you must miss a laboratory meeting, contact the instructor as soon as possible. Excused absences include documented illness, university-sponsored activities reported in advance, and documented emergencies. The instructor will determine an appropriate path based on the experiment, available equipment, team progress, and remaining laboratory schedule. An unexcused absence may result in no credit for the associated individual contribution and may affect the laboratory grade.

Collaboration, AI tools, and academic honesty

Collaboration is fundamental to this course. Team members should discuss the physics, divide experimental responsibilities, examine one another's calculations, and jointly revise the submitted notebook. Individual effort reports, however, must be completed independently.

Thoughtful use of outside resources is permitted. You may use textbooks, websites, and AI tools such as ChatGPT to clarify concepts, troubleshoot code, or evaluate reasoning. When assistance contributes materially to the notebook, acknowledge it briefly in the notebook. Every team member must understand, verify, and be able to explain the submitted measurements, calculations, code, figures, and conclusions. Copying work from another team or submitting unverified work produced by another person or tool violates this expectation even when the source is named.

The spirit and requirements of the Christopher Newport University Honor Code apply to all course work.

Investigating the Natural World

This course may fulfill part of the Investigating the Natural World requirement. Students who successfully complete the course should be able to:

- identify methods of inquiry that lead to scientific knowledge;
- distinguish science from pseudoscience;
- make predictions about natural phenomena using theories and models as unifying principles;
- analyze and interpret data in graphical, symbolic, and numerical forms;
- discriminate between association and causation; and
- identify types of evidence used to establish causation.

Official university syllabus statements — AY 2026–2027

Academic honesty

Academic honesty is fundamental to the mission of a university. Cheating, plagiarism, and any other violations of the Honor Code may result in academic remedies such as a failing grade on the assignment or the course (depending on the severity of the offense). Violations will also be reported to the Center for Honor Enrichment and Community Standards (CHECS).

The Christopher Newport University Honor Code:

“On my honor, I will maintain the highest standards of honesty, integrity and personal responsibility. This means I will not lie, cheat or steal, and as a member of this academic community, I am committed to creating an environment of respect and mutual trust.”

University statement on inclusion and belonging

The Christopher Newport University community engages and respects different viewpoints, understands the cultural and structural context in which those viewpoints emerge, and questions the development of our own perspectives and values, as these are among the fundamental tenets of a liberal arts education.

Accordingly, we affirm our commitment to a campus culture that embraces the full spectrum of human attributes, perspectives, and disciplines and offers every member of the University the opportunity to become their best self.

Understanding and respecting differences can best develop in a community where members learn, live, work, and serve among individuals with diverse worldviews, identities, and values. We are dedicated to upholding the dignity and worth of all members of this academic community, such that all may engage effectively and compassionately in a pluralistic society.

Questions, suggestions, or concerns regarding inclusion on campus may be directed to belonging@cnu.edu.

Disabilities and accessibility

In order for a student to receive an accommodation due to a disability, that disability must be on record in the Office of Student Affairs, 3rd Floor, David Student Union (DSU). If you have a diagnosed disability, please contact Accessibility & Care Team Support in Student Affairs (594-7160) or email careteam@cnu.edu to discuss your needs. You may visit <https://cnu.edu/life/acts/accessibility/> to request accommodations or view additional information.

In order to provide access to educational programs in the best manner possible, students with documented disabilities should notify the instructor at least seven days prior to the point at which they require an accommodation (the first day of class is recommended), in private, if an accommodation is needed. The instructor will provide students with disabilities with the reasonable accommodations approved and directed

by the Office of Accessibility and Care Team Support. Work completed before the student notifies the instructor of their disability may be counted toward the final grade at the sole discretion of the instructor.

Success

I want you to succeed in this course and at Christopher Newport. I encourage you to contact me during office hours or schedule an appointment to discuss course content or ask questions. If I become concerned about your course performance, attendance, engagement, or well-being, I will contact you first. I also may submit a referral through our Captains Care Program. The referral will be received by Student Affairs, the Center for Academic Success, and other departments, when appropriate. If you are an athlete, the Director of Student-Athlete Transition and Support will be notified. Someone from Student Affairs or the Center for Academic Success will contact you to set up a plan for success and connect you with the appropriate resources. Please remember that this is a means for me to support you and help foster your success at Christopher Newport.

Academic support

The Center for Academic Success (CAS) provides numerous free resources to help students succeed in their courses. These resources include individual peer mentoring, peer tutoring, and group/test review sessions. Academic Success Coordinators can meet with you regularly to discuss your plan to complete your coursework while balancing your extracurricular activities. They can help you develop individualized study skills, create time-management plans, and techniques to prepare for exams and other assignments. The Center also houses the Alice F. Randall Writing Center. Writing consultants can help you at any stage of the writing process, from invention to development of ideas to polishing a final draft.

You may contact the Center for Academic Success to request a tutor, meet with a peer mentor or writing consultant, or make an appointment to talk with a staff member about study skills and strategies. The Center is located in Christopher Newport Hall, first floor, room 124. You may email academicsuccess@cnu.edu, call (757) 594-7684, or visit <https://cnu.edu/center-for-academic-success>.

Public health

The university will provide guidance on public health issues and students will be expected to comply with university protocols.

Course materials

All content created and assembled by the faculty member and used in this course is to be considered intellectual property owned by the faculty member and Christopher Newport University. It is provided solely for the private use of the students currently enrolled in this course. To ensure the free and open discussion of ideas, students may not make any original course content, including but not limited to lectures, discussions, videos, handouts, and/or activities, available to anyone not currently enrolled in the course without the instructor's advance written permission. This means that students may not record, download, screenshot, or in any way copy original course material for the purpose of distribution beyond this course. A violation may be considered theft. It is the student's responsibility to protect course materials when accessing them outside the physical classroom.

Course-specific authorization: Materials intentionally published by the instructor in the public [brash99/Phys201L GitHub repository](#) are authorized for public access and use under the license included in that repository. This authorization does not extend to private feedback, LMS content, recordings, unpublished materials, or student information.